

RetailEdge AI — System Design Plan

Project	RetailEdge AI — Intelligent Retail Analytics System
Problem Statement	SIH 26179 (Qualcomm Inc) — AI-powered retail intelligence: real-time shopper analytics, automated inventory visibility, proactive queue management
Document Type	System Design Plan (v1.0)
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1. Executive Summary

RetailEdge AI is an edge-native retail intelligence platform that converts existing CCTV feeds into real-time, privacy-safe business insights using three integrated modules — **Shopper Analytics**, **Inventory Monitoring**, and **Queue Intelligence** — running entirely **on an in-store edge device**. No video ever leaves the premises; only anonymized, structured events are stored locally and optionally synced to a central cloud for multi-store BI.

One-line pitch: *Cameras + edge compute run all vision AI inside the store; only anonymized events (counts, alerts, KPIs) cross the network — no video, no faces, no cloud dependency.*

1.1 Why this design wins on this problem statement

Qualcomm's PS 26179 explicitly demands: "*maintaining privacy and minimizing cloud dependency*", "*operate even during internet disruptions*", "*AI inference should happen locally on the edge devices*", and "*convert video streams into actionable business insights.*" Most teams will build a cloud dashboard with recorded clips and violate the core constraint. This system is **offline-first by construction**: full functionality with zero internet, proven live in the demo by pulling the network cable.

1.2 Design principles

#	Principle	How the design achieves it
P1	Real-time	Inference latency < 100 ms on-device; alert-to-dashboard < 2 s end-to-end
P2	Privacy by architecture	No face recognition, no video retention, metadata-only storage & sync; kill-switch
P3	Offline-first	Zero network calls anywhere in the pipeline; store-and-forward sync is optional
P4	Low cost	₹18k-₹60k per store hardware (BOM in Appendix B); one box serves 4-8 cameras
P5	Works with what stores have	Ingests existing CCTV via RTSP; no POS/inventory system is a prerequisite
P6	Deployable day one	COCO-pretrained YOLO11n + heuristics — no per-store model training required for MVP
P7	Calibration, not retraining	Per-store tuning via <code>store_config.yaml</code> thresholds, never code changes
P8	Graceful degradation	Every module independent; one failing camera or module never takes down the demo or the store

1.3 Design sources

This plan consolidates four prior project artifacts: the full solution blueprint (`fusion-response 2026-09-09 (1).md`), the phased implementation plan (`fusion-response 2026-09-09.md`), the SIH MVP scope targeting the Jetson Orin Nano (`MVP-P0C-Plan.md`), and the market/technology research brief (`retail-intelligence-market-research.md`). Market figures cited in §14 are sourced in Appendix C.

2. Goals, Non-Goals & Success Metrics

2.1 Goals (MoSCoW: Must-have)

ID	Goal	Success Metric
G1	Entry/exit people counting with real camera streams	Entry/exit count MAE $\leq$ 2% (pilot footage), $\leq$ 5% (MVP demo)
G2	Zone dwell time + footfall heatmap	Heatmap spatially matches observed lingering; dwell error $\leq$ 15%
G3	Out-of-stock / low-stock detection	OOS precision $\geq$ 0.85 (MVP) / 0.90 with recall $\geq$ 0.85 (pilot target)
G4	Queue length measurement + congestion prediction	Predicted wait within $\pm 20\%$ of measured at 10-min horizon
G5	Full on-device inference (edge demo box)	$\geq$ 30 FPS per stream on Jetson Orin Nano (TensorRT INT8)
G6	Operates with zero internet	All modules functional with network cable pulled
G7 x	Single-store operator dashboard	KPIs, alerts, heatmap visible in one UI served from the edge box
G8 x	Privacy compliance by construction	No video written to disk; no PII in events or sync payload

2.2 Non-goals (explicitly out of scope — Won't-have)

- x Face recognition or any person re-identification across sessions
- x Per-SKU product recognition in MVP (heuristic shelf-gap detection instead; 2-stage model is the production upgrade, §17)
- Cloud video storage, cloud inference, or any real-time cloud dependency
- POS/ERP/Tally/Zoho/SAP integration in MVP (roadmap slide, §17)
- Fleet management, OTA updates, multi-store console (post-SIH)
- Autonomous checkout (adjacent market — AiFi/Trigo territory)

2.3 Should-have (post-MVP) & Could-have (roadmap)

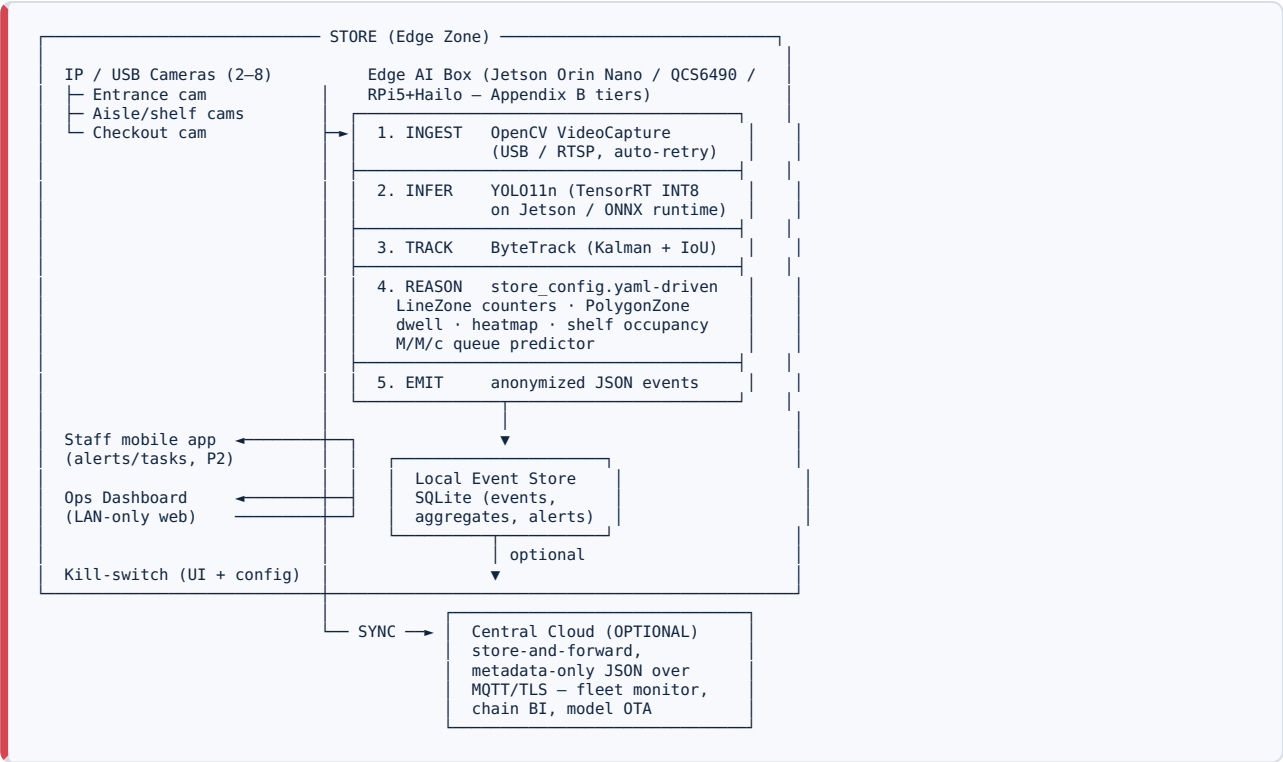
MoSCoW	Item
Should	Daily PDF/HTML analytics report generator; replay harness for recorded footage; staff mobile alert push
Could	Gradient-boosted footfall forecaster (CPU-only, offline); planogram compliance module; chain-level cloud BI

3. Users & Personas

Persona	Role in system	Primary jobs-to-be-done	Interface
Store owner/manager (kirana, supermarket, pharmacy)	Primary buyer; consumes KPIs & alerts	"Are we losing sales to stockouts? Do I need another counter at 6 pm?"	Dashboard (LAN browser / phone)
Floor staff / replenisher	Alert responder	"Where do I go right now and what's empty?"	Alert feed with aisle location + blurred snapshot
Chain operations lead	Fleet-level consumer (Phase 4)	"Which stores underperform on footfall conversion / OOS SLA?"	Central BI (cloud-optional)
Store IT / installer	Deployment & maintenance	"Install, calibrate, and leave it running without babysitting"	Config tool, systemd services, watchdog
The customer being observed	Never sees the system; must remain anonymous	—	Protected by design: no faces, no identity, no stored video

4. High-Level System Architecture

4.1 Component diagram (production form)



Key data flow: Camera frame → detect → track → business logic (zones/planogram/queue) → anonymized event → local DB → dashboard/alert. Raw frames are processed in memory and **never written to disk** (enforced by pipeline design, not policy — see §7).

4.2 Why each layer lives at the edge

Concern	Decision	Rationale
Inference	Edge box	PS requirement; < 100 ms latency; zero bandwidth cost (57 GB/day if cloud-video — see §4.3)
Event storage	Local SQLite	Full offline operation; single file, zero ops burden
Dashboard	LAN-served from the box	Judges/operators open it from any phone on store Wi-Fi — with no internet
Multi-store sync	Optional, store-and-forward, metadata-only	Internet outage never blocks the store; sync is a resume, not a dependency
Model updates	USB sideload / OTA when online	Offline-capable deployment lifecycle

4.3 Bandwidth feasibility (the number that justifies edge)

Cloud-video approach for 6 streams × 14 h/day at 1.5 Mbps: $6 \times 1.5 \text{ Mbps} \times 50,400 \text{ s} \div 8 \approx 57 \text{ GB/day}$ upload — impossible on Tier-2/3 links. The edge design sends **only events**: a full day of events for one store is < 5 MB ($\approx 0.009\%$ of the video payload).

5. Module Designs

5.1 Module M1 — Shopper Analytics (entrance/aisle cameras)

Feature	Approach	Output
Entry/exit counting	YOLO11n <code>person</code> class → ByteTrack IDs → LineZone crossing with direction test	In/out counters, net occupancy
Footfall trends	Hourly + day-part aggregation of line-cross events → <code>footfall</code> table	Trends by time/day/zone
Dwell time	Track persists inside PolygonZone → $\tau = t_{\text{exit}} - t_{\text{enter}}$, filtered for track fragmentation (< 3 s fragments dropped)	Per-zone dwell seconds
Heatmap	Accumulate Gaussian kernels at track centroids per zone (Supervision <code>HeatmapAnnotator</code>)	Live + hourly PNG heatmap

Heatmap accumulation per frame:

$$H(x, y) = \sum_{i \in \text{active tracks}} \exp(-\|p_i - (x, y)\|^2 / 2\sigma^2)$$

Design note: one wide-angle aisle camera covers 2-3 zones via polygon definitions — keeps camera count (and store cost) low.

5.2 Module M2 — Inventory Monitoring (shelf cameras)

MVP strategy (zero training): shelf-gap heuristic. Products are a huge, store-specific label space — per-SKU recognition needs ~50-100 images/SKU, which is why the MVP uses:

Stage	Method	Notes
Planogram definition	Slots as bboxes in <code>store_config.yaml</code> (drawn once per shelf from a reference frame)	Versioned per store
Facing occupancy	Per slot: edge density + background-color ratio (classic OpenCV)	Works for SKUs never trained on
Debounce	Alert only after N consecutive frames below threshold (N = 30)	Prevents shopper-occlusion false positives
Output	LOW_STOCK / OUT_OF_STOCK alert + blurred snapshot + aisle location → alert feed → staff task	Creates measurable replenishment SLA loop

$\text{occupancy}_{\text{slot}} = (\sum \text{detection areas in slot}) / (\text{slot area}) \rightarrow \text{alert if } < \theta_{\min} \text{ for } N \text{ consecutive frames}$

Production upgrade (post-SIH): 2-stage model — generic product detector (category-level, trains once) + per-store lightweight classifier (CLIP-style head). Scales across stores without per-SKU retraining.

Known hard problem: shopper blocking shelf → false OOS; lighting shifts; reflective packaging; planogram drift. Mitigations: aggressive debounce, per-clip threshold calibration, controlled shelf-camera angles, isolatable module (its failure never degrades M1/M3).

5.3 Module M3 — Queue Intelligence (checkout cameras)

Measurement: 1. Count heads per checkout queue polygon each frame; EMA-smooth: $\hat{q}_t = \alpha \cdot q_t + (1-\alpha) \cdot \hat{q}_{t-1}$, $\alpha \approx 0.3$ 2. Track queue-entry → billing-start → empirical service-time distribution; queue-entry → counter → wait time

Prediction (fully offline math): M/M/c queueing model with live arrival rate λ and measured service rate μ per counter:

$$\rho = \lambda / (c \cdot \mu) \quad \cdot \quad W_q \approx \lambda / (c \cdot \mu)^2 \cdot 1 / (1 - \rho) \quad (\text{valid for } \rho < 1; \text{ wait predicted at 10-min horizon})$$

Decision rule: predicted 10-min wait > threshold (default 5 min) → **"Open counter +1"** recommendation on dashboard.

Why M/M/c, not ML: it needs no training data, runs in microseconds, and is transparent to judges/operators ("λ from today's footfall, μ from measured service times"). A gradient-boosted λ forecaster is the optional upgrade (§17) — still CPU-only at the edge.

Engineering caution (known failure mode): the hard part is feeding M/M/c clean λ and μ from noisy vision data. Mitigations: EMA smoothing, minimum-queue-standstill detection, and per-store μ calibration from measured events.

5.4 Module M4 — Store Operations Dashboard

Streamlit app served from the edge box over LAN:

- **KPI cards:** live footfall, occupancy, OOS alerts, avg queue length, predicted wait, staff efficiency
- **Alert feed:** stock shortages + queue build-up, newest first, with blurred snapshots
- **Tabs:** Heatmap view · Shelf view (overlay + slot status) · Queue strip

- **Reports:** daily/weekly analytics summary (Should-have; simple HTML→PDF)
- **Privacy controls (always visible):** "Processing on-device — no video stored" badge, kill-switch toggle, current retention clock status

6. Edge AI Stack & Hardware Plan

6.1 Software stack (identical code on laptop and edge box)

Layer	Choice	Notes
Detection	Ultralytics YOLO11n → TensorRT INT8 engine (Jetson) / ONNX (others)	COCO-pretrained, <code>person</code> class; no training needed
Tracking / zones / heatmap	Supervision (ByteTrack, LineZone, PolygonZone, HeatmapAnnotator)	Same library everywhere — the port is painless
Queue math	M/M/c predictor (pure Python)	Deterministic, testable
Shelf OOS	Heuristic per planogram slot	Zero training data
Cameras	OpenCV <code>VideoCapture</code> (USB or RTSP)	One code path for webcam/phone/IP cam
Storage	SQLite	On the edge box
Dashboard	Streamlit	Served over LAN
Runtime (Jetson)	JetPack + TensorRT	DeepStream explicitly skipped — added complexity, zero demo value

The 10-line port trick: `YOLO("yolo11n.pt")` (laptop/PyTorch) → `YOLO("yolo11n.engine")` (Jetson/TensorRT INT8). Everything else is unchanged, so development proceeds on a laptop while hardware is sourced — and the laptop is a permanent fallback demo device.

6.2 Hardware tiers (per store; detailed BOM in Appendix B)

Tier	Hardware	AI capacity	Cameras	Cost	Best for
Entry	RPi 5 (8GB) + Hailo-8L (26 TOPS) + 64GB SSD	2–4 streams @ 10–15 FPS	2–3	₹18k–25k + cams	Kirana, small pharmacy
Sponsor-aligned	Qualcomm QCS6490 Edge AI Box (12 dense TOPS NPU, 30–50 AI threads)	5 concurrent cameras, 4K60 decode	5	₹30k–45k + cams	Supermarket — maps onto Qualcomm's own Insight Platform roadmap
Mid (demo tier)	Jetson Orin Nano 8GB + NVMe	6–8 streams, YOLO11n > 30 FPS/stream (TensorRT)	4–6	₹30k–45k + cams	Supermarket; the SIH demo device
Large	Jetson Orin NX or 2× Orin Nano	8–12 streams	8–12	₹60k+ + cams	Large-format

Sponsor-alignment rationale: this is a Qualcomm problem statement. The QCS6490 tier is the deployment reference, cited alongside Qualcomm's Insight Platform (Dec 2025, edge-first VSaaS

with exactly these retail analytics). The Jetson is the *demonstration* device; the Qualcomm tier is the *deployment* story — and ONNX export keeps the stack portable across both.

6.3 Performance budget (Jetson Orin Nano, TensorRT INT8)

Stage	Budget
Decode (4 streams, 720p15)	~1 ms/stream (NVDEC)
YOLO11n inference	~8-15 ms/batch
ByteTrack + zone logic	< 5 ms/frame
Event emit + DB write	< 1 ms
End-to-end latency target	< 100 ms; alerts < 2 s (includes debounce windows)

7. Privacy & DPDP Compliance by Design

7.1 Regulatory context

The **DPDP Act 2023 + DPDP Rules (finalized Nov 13, 2025)** explicitly cover AI-enabled CCTV. Full compliance is due **May 13, 2027**; penalties reach **₹250 Cr** for security-safeguard failures. Video of identifiable shoppers is personal data; there is no broad legitimate-interest escape hatch — consent + limited legitimate uses only.

7.2 Compliance by architecture (not policy)

Threat / obligation	Design answer
Identifiable video captured	No face recognition anywhere in the pipeline; person = anonymous track ID (<code>track_42</code>), never a person
Video retention	Frames processed in memory only — no video write path exists in the pipeline code (enforced by design, verified in tests)
Data minimization	Events contain counts/durations/zones only — no imagery, no coordinates linked to identity, no biometric templates
Snapshots for alerts	Blurred at generation time (face/regions masked)
Breach exposure	Nothing sensitive stored → nothing sensitive to breach; sync payload is JSON metadata only
Consent / notice	Store signage template + one-page privacy notice provided with the product kit
Retention clocks	Aggregates auto-pruned after configurable window (default 90 days)
Kill-switch	One toggle in the dashboard (and config flag): halts all cameras + wipes in-flight buffers — demonstrate it live

7.3 The privacy pitch beat

"Pull the network cable and the system keeps working. Open the database and every row is `{'event': 'queue_alert', 'est_wait_min': 6.2}` — no faces, no video, no PII. This is DPDP-compliant by construction, and it's the lowest-risk architecture available under the ₹250-Cr penalty regime."

Less data leaving the store = less DPDP exposure — privacy hardens the offline-first moat instead of competing with it.

8. Data Architecture

8.1 Event schema (v1 — the contract between pipeline, DB, and dashboard)

```
{
  "event_id": "evt_20260910_141233_00042",
  "store_id": "IN-BLR-014",
  "ts": "2026-09-10T14:12:33.512+05:30",
  "type": "entry_exit | dwell | low_stock | out_of_stock | queue_sample | queue_alert",
  "camera_id": "entrance_1",
  "payload": {
    "zone_id": "promo_endcap_A",
    "count": 3,
    "dwell_sec": 47.2,
    "slot_id": "shelf_a3#12",
    "occupancy": 0.11,
    "est_wait_min": 6.2,
    "recommendation": "open_counter_plus_1"
  },
  "privacy": { "video_retained": false, "pii": false }
}
```

8.2 Local storage model (SQLite, on the edge box)

Table	Grain	Purpose	Retention
events	One row per emitted event	Audit trail, replay, debugging	30 days
footfall	(store, zone, hour)	Entry/exit & occupancy aggregates	90 days
dwell_stats	(zone, hour)	Mean/median dwell per zone	90 days
shelf_status	(slot, ts)	Occupancy samples + alert timestamps	90 days
queue_stats	(counter, minute)	Length, wait, service-time samples	90 days
alerts	One row per alert	Alert feed + SLA (ack → resolved)	90 days
snapshots	Blurred PNG paths	Alert evidence only — never raw video	7 days

8.3 Sync model (optional, store-and-forward)

- Queue: local outbox table → `mqtt/tls` publisher; exponential backoff; **causal ordering per store**
- Idempotency: `event_id` is the dedup key at the cloud sink

- Outage behavior: outbox grows on local disk (≈ 5 MB/day), sync resumes from cursor — **the store never waits for the cloud**
- Payload: JSON events only (§7); video never synced by design

9. APIs & Interfaces

9.1 Internal (process-internal interfaces, MVP)

Interface	Type	Contract
Pipeline → DB	Function call / queue	Event JSON v1 (§8.1)
DB → Dashboard	Streamlit read queries	Tables §8.2
Pipeline ↔ <code>store_config.yaml</code>	File (versioned per store)	Schema below

```
# store_config.yaml – the contract between modules; calibration-not-retraining lives here
store_id: "IN-BLR-014"
cameras:
  - { id: entrance_1, rtsp: "rtsp://192.168.1.10/ch1", role: entrance }
  - { id: shelf_a3,   rtsp: "rtsp://192.168.1.14/ch1", role: shelf }
  - { id: checkout,  rtsp: "rtsp://192.168.1.17/ch1", role: queue }
lines:
  - { id: entry_line, a: [100, 400], b: [900, 400], in_direction: down }
zones:
  - id: promo_endcap_A
    polygon: [[210, 300], [520, 300], [520, 640], [210, 640]]
    dwell_alert_sec: 45
planogram:
  shelf_a3:
    slots:
      - { sku: "SKU-88213", bbox: [40, 120, 180, 260] }
      - { sku: "SKU-90441", bbox: [190, 120, 330, 260] }
thresholds:
  out_of_stock_frames: 30
  queue_wait_min: 5.0
  occupancy_warn: 80
```

9.2 External (Phase 3+ — post-MVP roadmap)

API	Consumer	Notes
REST <code>/events</code> , <code>/kpis</code> , <code>/alerts</code>	Chain BI cloud, future integrations	Read-only, local network or sync channel
Webhook <code>/alerts</code>	Staff mobile app / WhatsApp bot (roadmap)	Debounced
Connector adapters (Tally, Zoho, SAP)	POS/ERP	Roadmap slide — not MVP

10. Deployment Architecture

10.1 SIH MVP deployment (the demo)

```
Laptop (dev, days 1-8) —same code—> Jetson Orin Nano (days 9-12)
      |
      | permanent fallback demo <—
      | (swap yolol1n.pt ↔ yolol1n.engine; everything else identical)
      |
USB webcam(s) &#8594; Jetson &#8594; pipeline (detect/track/reason/emit) &#8594; SQLite &#8594; Streamlit (LAN)
Judges' phones/browsers on store LAN &#8594; dashboard — no internet anywhere in the loop
```

10.2 Pilot deployment (3-5 stores)

- Pre-imaged SD/NVMe flash with systemd services (`ingest.service` , `pipeline.service` , `dashboard.service`) + watchdog auto-restart
- Camera placement per site survey (lighting day/night, mounting, FOV sketches)
- On-site calibration: draw lines/zones/planogram once → `store_config.yaml`
- Shadow mode first (all alerts logged, none pushed) → measured accuracy → live alerts
- Dashboard on store LAN; sync agent optional and off by default

10.3 Fleet deployment (Phase 4+)

- Per-store edge box identical; central cloud is **optional BI**, never a runtime dependency
- Model updates: signed model bundles via OTA when online, USB sideload when not

11. Reliability, Observability & Failure Modes

11.1 Failure-mode matrix

Failure	Detection	Automated response	Degraded behavior
Camera stream drops	Frame-gap watchdog per camera	RTSP reconnect w/ backoff	Other cameras unaffected; dashboard shows camera-down chip
Model inference stall	Heartbeat between frames	Process restart (systemd), engine reload	Brief gap; counters resume from tracks
Disk full	Inode/space monitor	Rotate/prune per §8.2 retention	Events prioritized over snapshots
Edge box crash/ power cut	systemd, watchdog timer	Auto-restart all services	Store reopens dashboard in < 60 s; no data beyond crash lost (WAL mode)
Internet outage	Sync agent fails	Buffer in outbox	Zero functional impact — the core promise
Thermal throttle (Jetson)	<code>nvpmodel</code> /temp telemetry	Drop to 15 W mode; process every 2nd frame	FPS halves, still demoable
Zone mis-calibration (drift)	Accuracy monitor vs shadow audit	Flag for recalibration	Conservative thresholds until recalibrated

11.2 Observability (local-first)

- Per-stream FPS/latency counters exported to local Prometheus + Grafana (Phase 3+; MVP: dashboard status chips)
- Daily self-test: fake frame through full pipeline → expected event → alert on silence
- Chaos testing (Phase 3 gate): kill camera / fill disk / kill process / pull network → all recover < 5 min

12. Phased Execution Plan

12.1 Full program (5 phases, 30+ weeks)

Phase	Weeks	Name	Goal	Hard Exit Gate (evidence, not opinion)
P0	1–4	Discovery & Foundation	Requirements frozen, pilot site secured, baseline model proven on real footage	Baseline person-detector mAP@50 $\geq$ 0.85 on real store footage; <code>PRD.md</code> , <code>PRIVACY.md</code> , site survey, dataset v0
P1	5–12	Bench MVP	All 3 modules end-to-end on edge hardware in lab, dashboard v1	3 streams $\times$ 15 FPS sustained 24 h; entry/exit MAE $\leq$ 5%; Internal Demo Day
P2	13–22	Pilot Deployment	3–5 live stores, shadow → live alerts, staff trained	Count MAE $\leq$ 2%; OOS P/R $\geq$ 0.90/0.85; uptime $\geq$ 98%; weekly shadow audits
P3	23–30	Hardening & Release 1.0	Fault-proof, secure, fleet-managed, technician-installable	72 h soak clean; all chaos faults recover < 5 min; OTA 100%
P4	31+	Scale & Commercialize	10–20 stores, partner channel, chain BI	Unit economics proven; NPS $\geq$ 40 from store managers

12.2 SIH MVP timeline (~2 weeks — the buildable subset)

Days	Milestone	Exit check
0	Jetson borrow/PO in motion; repo + venv + <code>ultralytics/supervision/opencv/streamlit</code> ; hello-world person count on laptop webcam	Stack proven in one command
1-2	People counting + entry/exit LineZone (laptop webcam)	Counts a walker within ± 1
3-4	Dwell + heatmap on entrance clip	Heatmap matches lingering spots
5-7	Queue predictor (M/M/c) + shelf OOS heuristic on clips	Alerts fire on scripted footage
8	Streamlit dashboard + privacy controls (badge, kill-switch)	All 3 modules in one UI
9-10	Jetson bring-up: JetPack, <code>yolo export format=engine</code> , camera sanity	Engine loads, ≥ 30 FPS, outputs match laptop
11-12	Full demo on the Jetson ; record real-store clips	5-min run-through on the edge box, no laptop
13+	Rehearse narrative (offline cable-pull, kill-switch, metadata JSON) + laptop fallback test	Demo works on EITHER device

12.3 Demo narrative (10 minutes, moat-proofs built in)

- (1 min) Problem with sourced numbers: ~8% stockout rate, ~4% of revenue lost, ₹ hardware tiers
- (4 min) Live modules on the Jetson: counting reacting to people, shelf alert on scripted clip, queue "open counter +1"
- (4 min) **Moat proofs:** pull the network cable → keeps working; open the DB → metadata-only JSON; kill-switch → cameras halt; ROI slide → every alert maps to a ₹ number
- (1 min) Close: "Runs on QCS6490-class boxes today; designed alongside Qualcomm's own Insight Platform direction."

Contingencies (rehearsed, not improvised): Jetson absent → laptop runs identical code; no store clips → teammates act as shoppers on webcam; thermal throttle → 15 W mode or every-2nd-frame.

13. Team, Cadence & Governance

13.1 Core team (7-9)

Role	Count	Responsibility	Phases
Product / Field Lead	1	Requirements, store partnerships, pilot ops, ROI story	All
CV/ML Lead	1	Model architecture, training strategy, accuracy sign-off	All
ML Engineer	1	Training, annotation ops, quantization, model registry	P0-P3
Edge Engineer	1	Pipelines, Jetson/Hailo deployment, watchdogs	P1-P3
Backend Engineer	1	Event bus, DB, APIs, sync agent, integrations	P1-P4
Frontend Engineer	1	Dashboard, staff app, planogram tool	P1-P4
QA / Field Ops	1	Test protocols, ground-truth audits, chaos testing, installs	P2-P4
Data Annotators	2 PT	Bounding boxes, SKU labels, audits	P0-P2

13.2 Cadence

- **Daily:** 15-min standup per workstream
- **Weekly:** cross-team demo (working software only) + metrics review
- **Phase gates:** formal review at close; exit criteria checked against evidence (test logs, audit reports). *A phase closes on data, not opinion.*
- **Tooling:** GitHub + Actions CI; W&B/MLflow registry; CVAT/Label Studio; Docker Compose; Prometheus + Grafana (local)

14. Business Case & Commercial Model

14.1 The problem economics (sourced — Appendix C)

Problem	Benchmark	Source basis
Stockouts	~8% average rate; ~4.1% of revenue lost; ~\$1.2T/yr global lost sales	NetSuite, Retail Wire, Mirakl (2025–26)
Shrinkage	~1.4–1.6% of sales; ~\$132B global (2024)	NRF benchmark, Building Security
Queue wait	Direct conversion impact; vendor CV claims: 45% shorter waits (treat as marketing)	Category claims
India TAM	Retail ~\$1.09T (2025) → \$2.36T (2030); 12–15M kiranas; organized only ~18–20%	IBEF, Cornell, Invest India
Market size	In-store analytics \$5.3B (2025) → ~\$16B (2031); CV-in-retail → \$12.6B (2033)	Mordor, Grand View (estimates — quote ranges)

→ The plan's targets (OOS ↓ ≥ 30%, wait ↓ ≥ 25%) are conservative against these benchmarks — credible to judges and store owners.

14.2 Pricing & unit economics (directional)

Tier	Hardware	SaaS (per store/mo)	Target segment
Kirana (1-2 cams)	₹18-25k	₹999	Light module set: footfall + queue
Supermarket (4-6 cams)	₹30-45k	₹2,499	Full 3 modules + reports
Chain (8+ cams / multi-store)	₹60k+	₹5,999+	Fleet BI, SLA dashboards, integrations

Rough ROI story per supermarket: 4.1% revenue exposure to OOS → 30% reduction = **~1.2% of revenue recovered** against a ₹2,499/mo software cost — the hardware pays back in months, not years. (Present as a worked example with pilot data, not a promise.)

14.3 Go-to-market sequence

1. **Modern trade / small supermarkets first** (budget fits, workflows match, staff present) — 3-5 pilot stores
2. Pharmacy chains next (planogram-heavy, high SKU churn)
3. Kirana last (price-sensitive; share-hardware or subsidized model; footfall+queue light SKU)

15. Risks & Mitigations

#	Risk	LxI	Mitigation	Owner
R1	Edge accuracy under Indian retail conditions (low light, crowds, occlusion, mannequins) — the #1 technical risk	HxH	IR-capable cameras in placement; hard-example mining; per-store threshold tuning; honest targets + shadow-mode validation before live alerts	CV Lead
R2	Shelf OOS false positives (shoppers blocking shelf, reflections, planogram drift)	HxM	30-frame debounce; per-slot calibration; isolated module; controlled shelf angles; demo on scripted clip	Edge Eng
R3	Jetson doesn't arrive / env setup friction on borrowed hardware	MxH	Laptop fallback is built into the plan (identical code); order/borrow early (Day 0 action)	Product
R4	No real store footage	MxH	Permission letter for a local store in Week 0; teammates-as-shoppers fallback; public retail datasets	Field Lead
R5	M/M/c polluted by noisy λ/μ from vision	MxM	EMA smoothing; standstill detection; μ calibration from measured events; conservative recommendation thresholds	ML Eng
R6	Judge tests the offline claim and something silently phones home	LxH	Zero network calls by construction; demo checklist grep of egress; rehearsed cable-pull beat	Edge Eng
R7	DPDP regime evolves before 2027 enforcement	LxM	Architecture is already minimal-exposure; monitor consent-manager framework (Nov 2026); legal review pre-pilot	Product
R8	Well-funded competitors (Trax \$1.07B) enter India offline-first niche	MxM	Speed-to-pilot, store-owner references, price moat (\$16)	Product
R9	Analyst market numbers vary widely → credibility hit if quoted as fact	LxL	Quote ranges + sources; use per-retailer percentages for defensibility	Product
R10	Kirana price sensitivity limits TAM	HxL	Modern-trade-first GTM (\$14.3); light kirana SKU	Product

16. Competitive Positioning & Moat

16.1 Landscape

Company	Focus	Funding/traction	Gap vs us
Trax	Shelf/planogram recognition	~\$1.07B raised	Cloud-centric, enterprise-priced, Western
Focal Systems	Edge shelf cameras	~\$28-42M	Edge but US-market, shelf-only
AiFi / Trigo / Zippin	Autonomous checkout	\$65M+ (AiFi)	Adjacent market
Nodelflux	On-prem people counting	Indonesia	SE-Asia, security-centric
Wesense.ai	CCTV analytics	India	Cloud-assisted; no offline-first proof

The gap nobody occupies: offline edge inference + DPDP-native privacy + Indian price point + all three modules in one box.

16.2 The moat stack (in hackathon terms: what we can prove in 10 minutes)

#	Moat	Proof in the demo
	Offline-first, proven live	Pull the network cable; system keeps counting
	Privacy by architecture	Kill-switch click; no-video-written badge; show the metadata JSON
	Business-insight closure	ROI slide: every alert maps to a ₹ number; OOS ↓ 30% / wait ↓ 25% targets
4	India-depth	Works with existing CCTV; no POS prerequisite (the camera is the inventory system); Tier-2/3 pricing
5	Qualcomm alignment	QCS6490 reference tier; ONNX portability; Insight Platform narrative

16.3 What we deliberately do NOT compete on

- **Model accuracy races** — everyone has YOLO; it's a race to the middle
- **"We trained our own model"** — zero-training day-one deployment is the better story; fine-tuning is an option, not an identity

17. Roadmap & Future Enhancements

Horizon	Item	Notes
Post-MVP	2-stage product detector + per-store classifier	Replaces shelf heuristic; ~50-100 img/SKU via staff phone app
Post-MVP	Planogram compliance module	Misplaced-SKU flags from detected centroids vs grid
Post-MVP	Staff mobile app	Alert push + task completion → measurable replenishment SLA
Phase 3	Fleet console + OTA	Signed model bundles; multi-store health
Phase 4	Chain-level BI + POS connectors (Tally/Zoho/SAP)	Cross-store conversion, inventory reconciliation
Optional	Gradient-boosted λ forecaster	Rush-hour refinement of queue predictions; CPU-only at edge
Optional	Fine-tuned YOLO11n per archetype	Only if real footage shows pretrained misses; 1-2k frames, free Colab GPU, hours of work

Appendix A — Key Formulas

Quantity	Formula	Notes
Heatmap accumulation	$H(x,y) = \sum_i \exp(-\ p_i - (x,y)\ ^2 / 2\sigma^2)$	Gaussian kernels at track centroids
Zone dwell	$\tau = t_{\text{exit}} - t_{\text{enter}}$	Drop fragments < 3 s
Slot occupancy	$\sum \text{detection areas in slot} / \text{slot area}$	Alert if < θ_{min} for N frames
Queue smoothing	$\hat{Q}t = \alpha \cdot Qt + (1-\alpha) \cdot \hat{Q}t-1$	$\alpha \approx 0.3$
M/M/c utilization	$\rho = \lambda / (c \cdot \mu)$	System unstable if $\rho \geq 1$
M/M/c expected wait	$Wq \approx \lambda / (c\mu)^2 \cdot 1/(1-\rho)$	For $\rho < 1$; 10-min horizon
Recommendation rule	$Wq(t+10) > 5 \text{ min} \ \&\#8594; \text{open counter} + 1$	Threshold in <code>store_config.yaml</code>
Cloud-video bandwidth	$\text{streams} \ \&\#215; \text{Mbps} \ \&\#215; \text{seconds} \div 8$	$6 \times 1.5 \times 50400 \div 8 \approx 57 \text{ GB/day}$ — why edge wins
Edge event payload	< 5 MB/store/day	$\approx 0.009\%$ of video payload

Appendix B — Hardware BOM (per store)

Tier	Core	Add-ons	Cameras	Indicative cost
Entry (kirana/ pharmacy)	RPi 5 8GB + Hailo-8L (26 TOPS) + 64GB SSD	PSU, case, mounts	2-3 × 2MP PoE	₹18k-25k + cams
Sponsor-aligned (supermarket)	Qualcomm QCS6490 Edge AI Box (12 TOPS NPU; VVDN / Inventec / OnLogic builds)	NVMe	5 × 2-4MP PoE	₹30k-45k + cams
Mid / SIH demo	Jetson Orin Nano 8GB dev kit + NVMe	—	4-6 (demo: USB webcam ×2)	₹30k-45k + cams
Large	Jetson Orin NX or 2× Orin Nano	NVMe ×2	8-12	₹60k+ + cams

Cameras spec: 2MP-4MP PoE, H.264/H.265 with sub-stream (720p/15fps) to analytics; full-res stream only to NVR if one exists — **never to our box's disk**.

Appendix C — References (full list in research brief)

Market: Grand View Research (CV-in-retail 2025-33); Mordor Intelligence (In-Store Analytics, Aug 2026); DataIntel; Fortune Business Insights; Research and Markets (2026). **India:** IBEF Retail; Cornell SC Johnson (May 2026, 13M kiranas); Invest India; ACR Journal (UPI adoption). **Problem economics:** Mirakl (\$1.2T OOS); OpenSend; NetSuite (stockouts); vndly (4.1% revenue); NRF 2025; Building Security (\$132B shrink); invue (1.4-1.6%). **Competition:** Trax (\$640M Series E, traxretail.com); Bouncewatch (Focal vs Trax); TechCrunch (AiFi); StartUs (incl. Wesense.ai); nodeflux.ai. **Hardware/Qualcomm:** qualcomm.com/edge-ai-box; Qualcomm Insight Platform (Dec 2025 announcement + Retail PDF); Silex (QCS6490); Qualcomm IoT/Dragonwing; Hailo community + Ultralytics forum benchmarks (RPi5+Hailo, Jetson). **Privacy:** DPDP Act 2023; DPDP Rules (Nov 13, 2025) via Fisher Phillips (Feb 2026) and Linklaters; AI-enabled CCTV coverage confirmation (2025). Complete URLs: [retail-intelligence-market-research.md](#) §6.

End of System Design Plan v1.0 — RetailEdge AI (SIH 26179). Build artifacts referenced: solution blueprint, phased implementation plan, MVP/POC plan (Jetson), market research brief.